

BPM 1: Lecture 39

Flipped Classroom

REQUIRED READING material for this lecture are:

1. Watch this Video Recording for Flipped Classroom: Link to the video:
<https://sgu1.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=c99731bf-c699-41e4-8264-b34e00d4756a>

2. Read the following related REQUIRED pre-read material in the recommended Physiology textbook, chapter 6-Autonomic Nervous System, Under heading, “AUTONOMIC INTEGRATION”. Use link below to reach here:
<https://premiumbasicsciences-lwwhealthlibrary-com.sgu.idm.oclc.org/content.aspx?sectionid=253086161&bookid=3211>

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LEARNING OBJECTIVES

- **SOM.MK.II.BPM1.1.FTM.3.PHYS.0046** Explain the responses of end organs to stimulation of sympathetic and parasympathetic divisions of the autonomic nervous system.
- **SOM.MK.II.BPM1.1.FTM.3.PHYS.0047** List and explain the functions of the eye regulated by the autonomic nervous system, including pupillary constriction and dilation, accommodation, aqueous humor production, and upper eyelid elevation.
- **SOM.MK.II.BPM1.1.FTM.3.PHYS.0048** List the functions, afferent and efferent pathways and stimuli of the baroreflex. Predict clinical implications of baroreflex dysfunction.

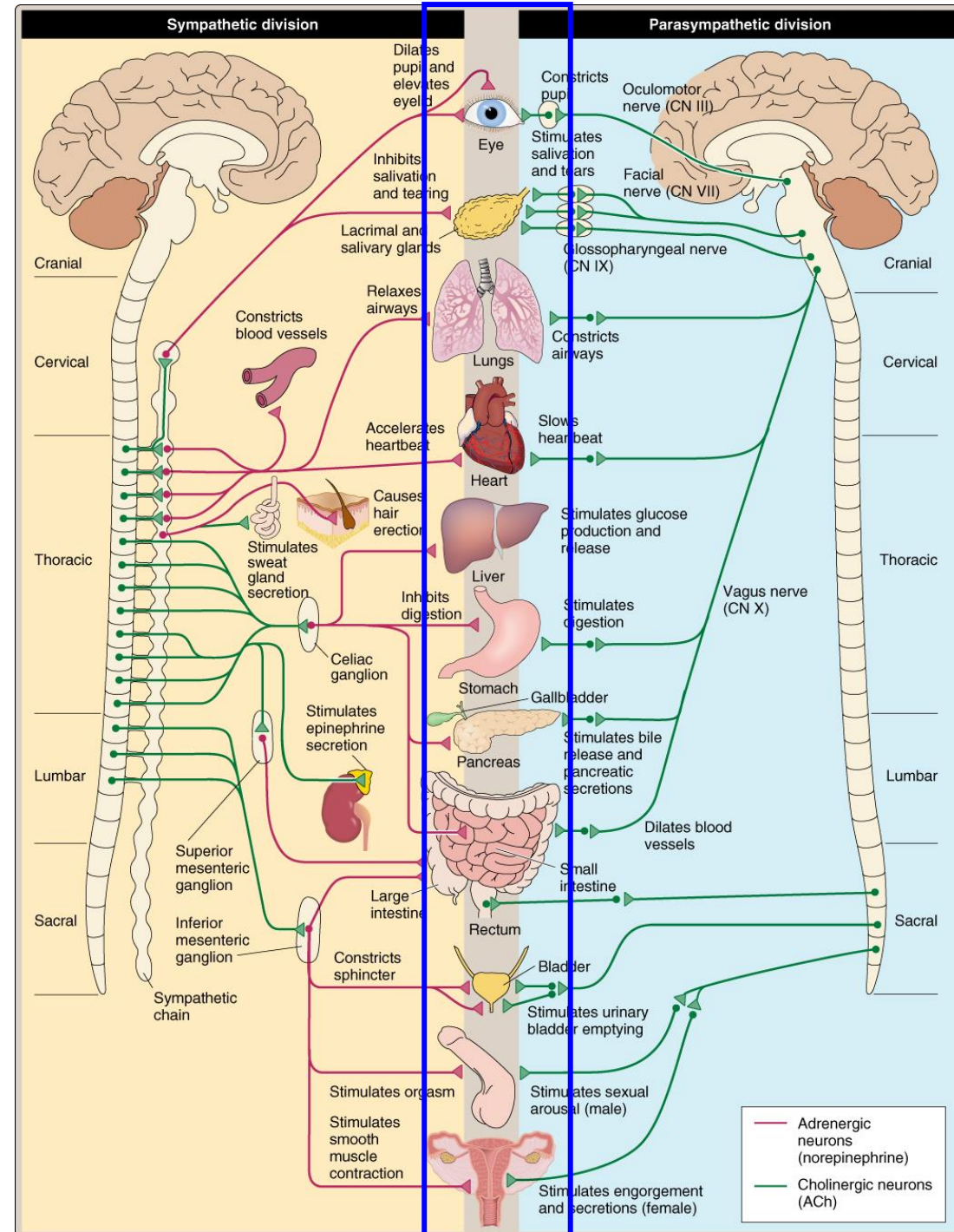
Read the following related REQUIRED pre-read material in the recommended Physiology textbook, chapter 6-Autonomic Nervous System,

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Dual innervation and antagonistic effects of ANS

There is an
equilibrium between
sympathetic and
parasympathetic
effects



List target organs that
has **only sympathetic**
innervation:

Modified from Bear, M.F.,
Connors, B.W., and Paradiso,
M.A. Neuroscience—Exploring
the Brain. Third Edition.
Lippincott Williams & Wilkins,
2007

End organ responses to sympathetic and parasympathetic divisions

Region	Effector	Parasympathetic <i>Muscarinic Receptors</i>	Sympathetic <i>Adrenergic Receptors</i>
Skin	Sweat glands	None	Secretion (cholinergic muscarinic receptors)
	Arrector pili muscles	None	Contraction α_1
Blood vessels	Skin (general)	None	Constriction α_1
	Skeletal muscle	None	Dilation β_2
	Viscera	None	Constriction α_1
Heart	SA Node	Decrease heart rate M2	Increase heart rate β_1
	AV Node	Decrease conduction velocity M2	Increase conduction velocity β_1
	Atria	Decrease contractility M2	Increase contractility β_1
	Ventricle	None	Increase contractility β_1

End organ responses to sympathetic and parasympathetic divisions

Region	Effector	Parasympathetic <i>Muscarinic Receptors</i>	Sympathetic <i>Adrenergic Receptors</i>
Lungs	Bronchioles	Constriction	Dilation β_2
	Glands	Secretion	Decrease
Gastrointestinal tract	Wall muscles	Contraction M3	Relaxation
	Sphincters	Relaxation	Contraction
	Glands	Stimulation M3	Inhibition
Metabolic functions	Liver	None	Glycogen breakdown and new glucose production
	Pancreas (insulin)	Increased secretion	Decreased secretion α_2
	Adipose Cells	None	Lipolysis β_3
	Kidney	None	Renin release β_1
	Adrenal medulla	None	Secretion of catecholamines (cholinergic nicotinic)

End organ responses to sympathetic and parasympathetic divisions

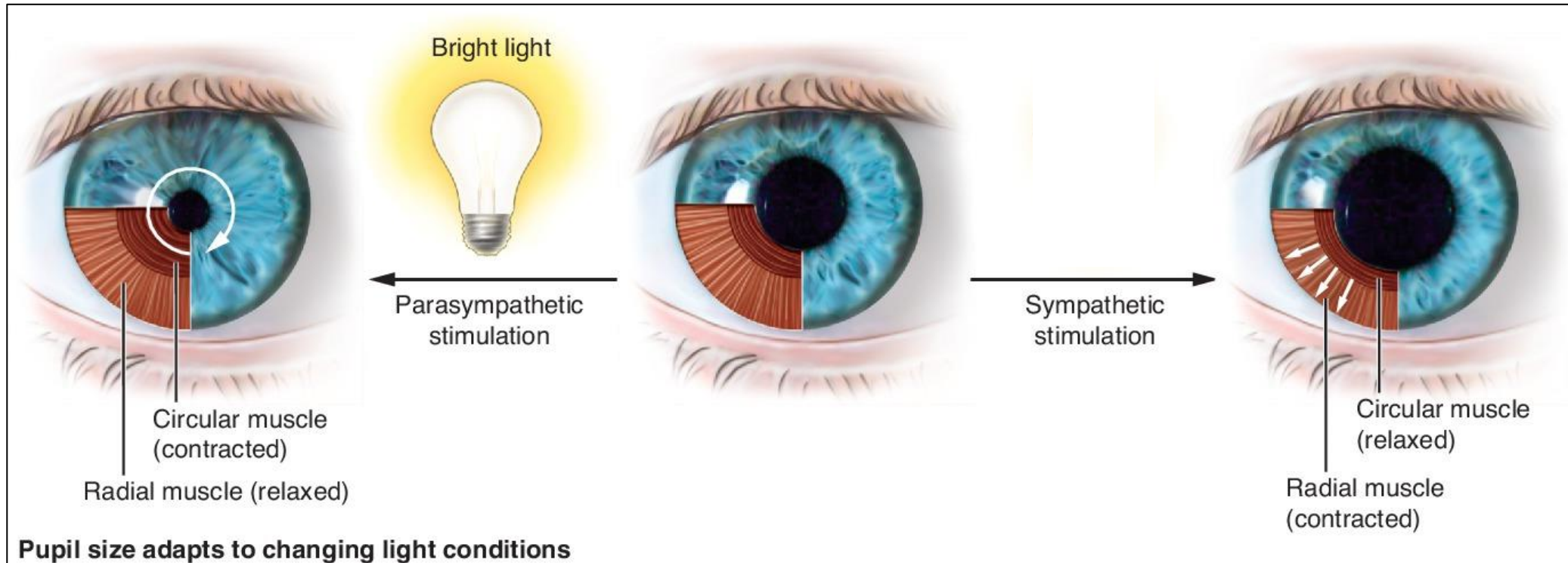
Region	Effector	Parasympathetic <i>Muscarinic Receptors</i>	Sympathetic <i>Adrenergic Receptors</i>
Urinary system	Detrusor	Contraction M3	Relaxation
	Sphincter	Relaxation	Contraction $\alpha 1$
Reproductive system	Uterus	Variable	Relaxation β_2
	Genitalia	Erection (ACh & NO) M3	Ejaculation
Eye	Circular muscle	Contraction M3	<i>None</i>
	Ciliary muscle	Contraction M3	<i>None</i>
	Radial muscle	None	<i>Contraction $\alpha 1$</i>
	Ciliary epithelium	None	<i>Secretion of aqueous humor $\beta_{1,2}$</i>
	Eyelid muscle (superior tarsal muscle)	None	<i>Elevates upper eyelid</i>

Circular muscle is also known as Sphincter pupillae muscle;
Radial muscle is also known as Dilator pupillae.

LEARNING OBJECTIVES

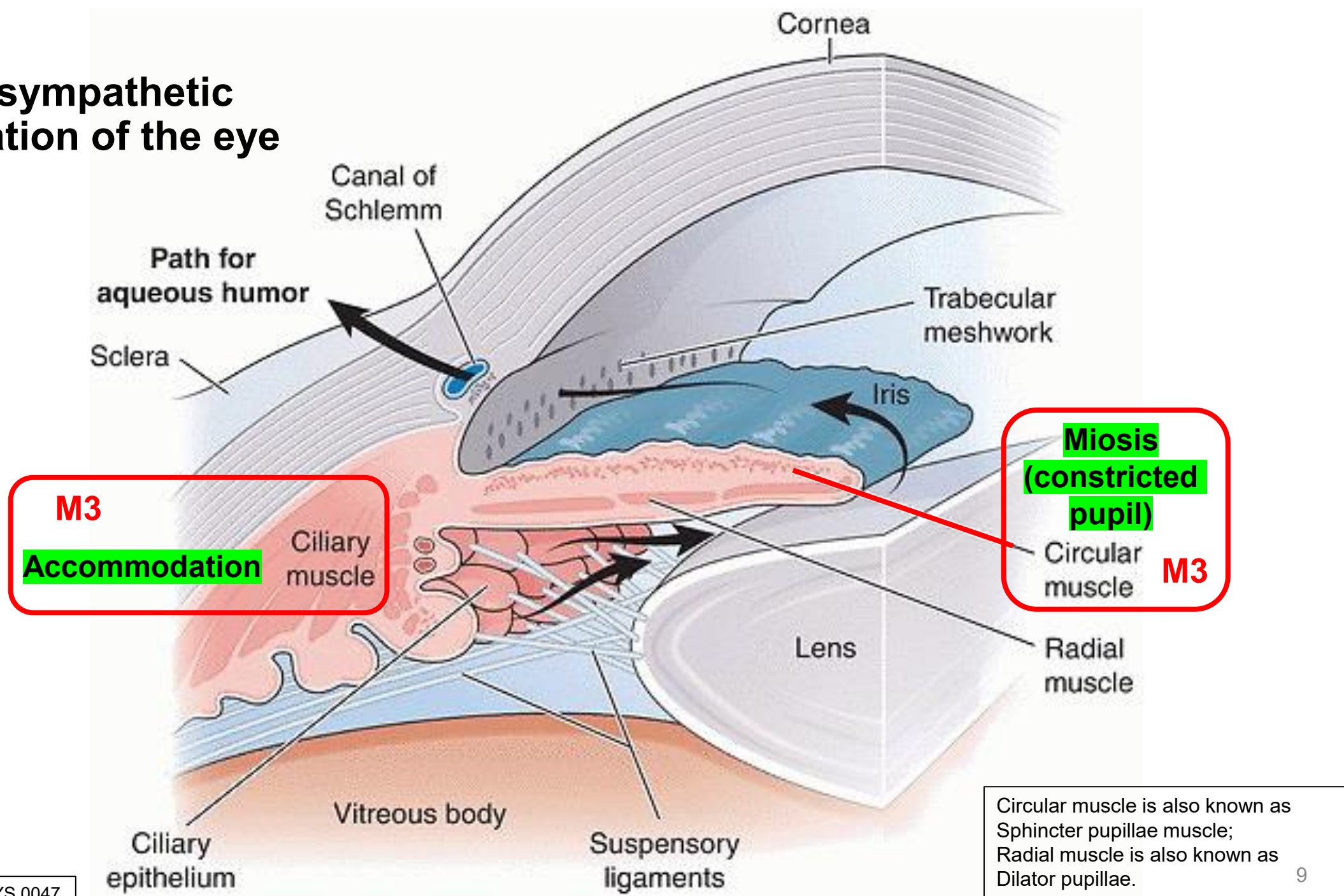
- **SOM.MK.II.BPM1.1.FTM.3.PHYS.0047** List and explain the functions of the eye regulated by the autonomic nervous system, including pupillary constriction and dilation, accommodation, aqueous humor production, and upper eyelid elevation.

Autonomic regulation of pupil light adaptation

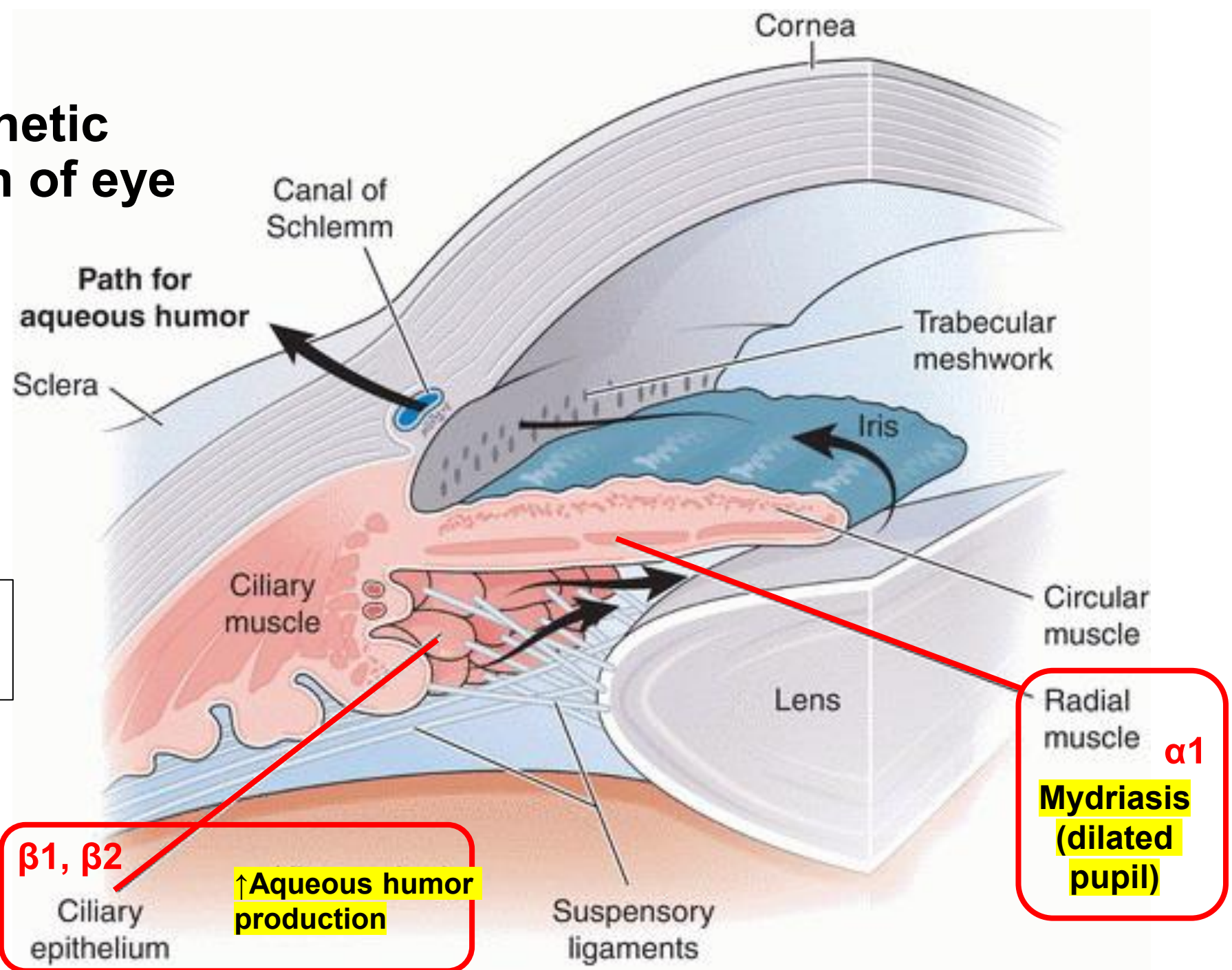


Circular muscle is also known as Sphincter pupillae muscle;
Radial muscle is also known as Dilator pupillae.

Parasympathetic Innervation of the eye



Sympathetic innervation of eye



Circular muscle is also known as Sphincter pupillae muscle; Radial muscle is also known as Dilator pupillae.

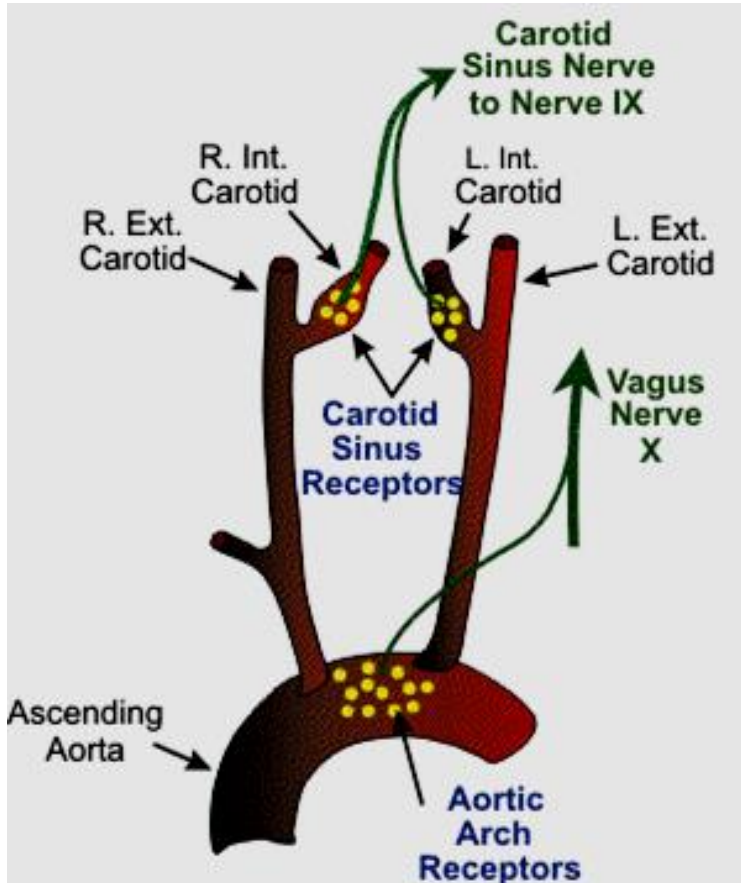
A 67-year-old man comes to the physician because of a 1-week history of blurry vision in his right eye. He has a 20-year history of type 2 diabetes mellitus. His pulse is 78/min and blood pressure is 158/92 mm Hg. Physical examination of the eye shows loss of pupillary light reflex in both eyes. He is diagnosed with dysfunction of the parasympathetic nerves supplying to the eyes. Dysfunction of which of the following structures in the eye is most likely in this patient?

- A. Dilator pupillae muscle (radial muscle)
- ✓ B. Sphincter pupillae muscle (circular muscle)
- C. Ciliary muscle
- D. Canal of Schlemm
- E. Ciliary epithelium
- F. Suspensory ligaments

LEARNING OBJECTIVES

- **SOM.MK.II.BPM1.1.FTM.3.PHYS.0048** List the functions, afferent and efferent pathways and stimuli of the baroreflex. Predict clinical implications of baroreflex dysfunction.

Arterial Baroreceptors initiate Baroreflex to regulate BP



Increase in Blood Pressure (**stimulus**)



Stretching of baroreceptors (**Sensors**)



Increased action potentials in the afferent nerves connected to the baroreceptors (**Afferent n**)

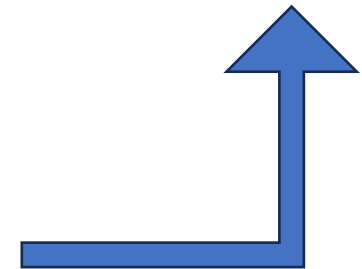


Cardiovascular regulatory centers in brain stem (**Center**)

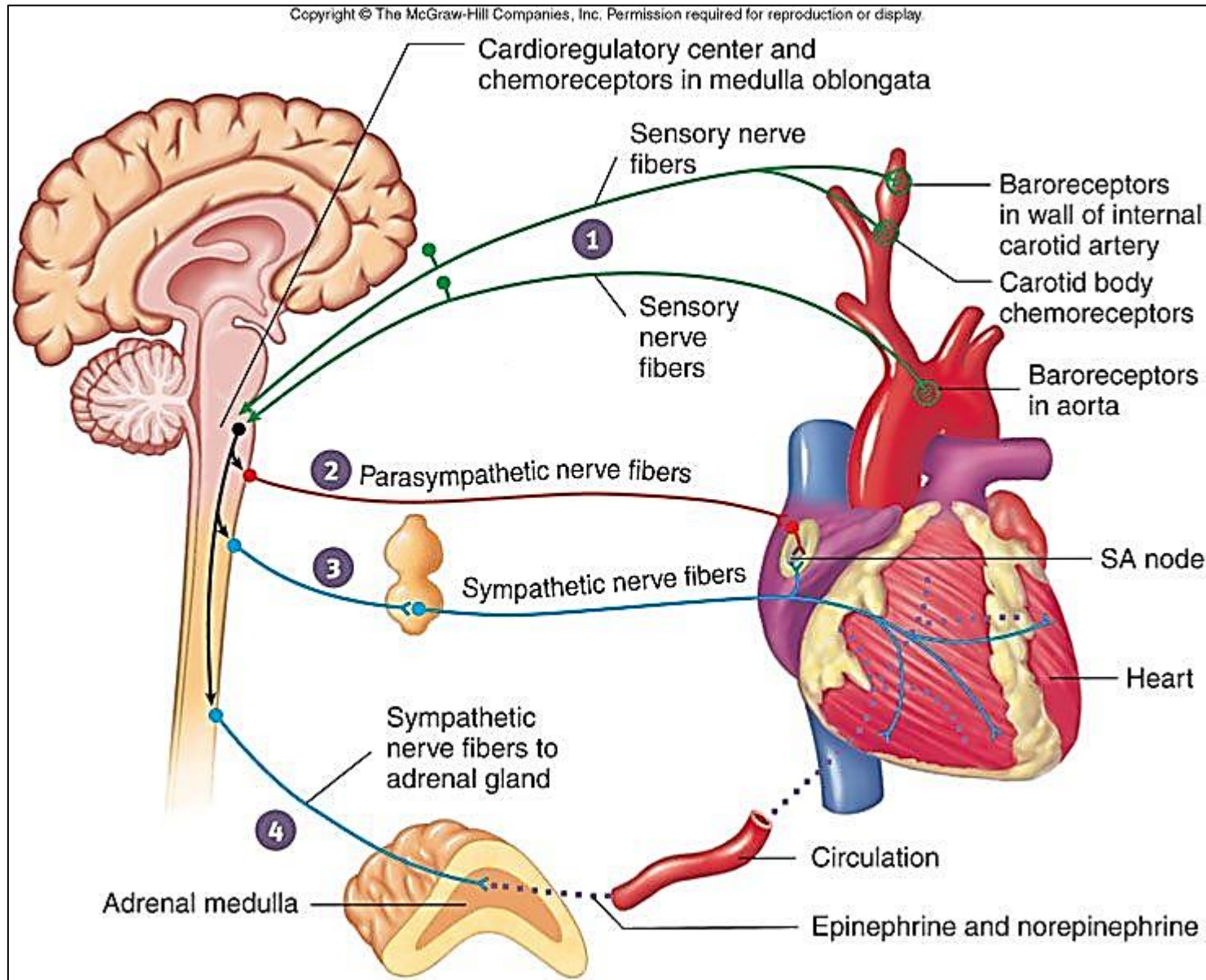


Influence on Autonomic NS (**Efferent n**)

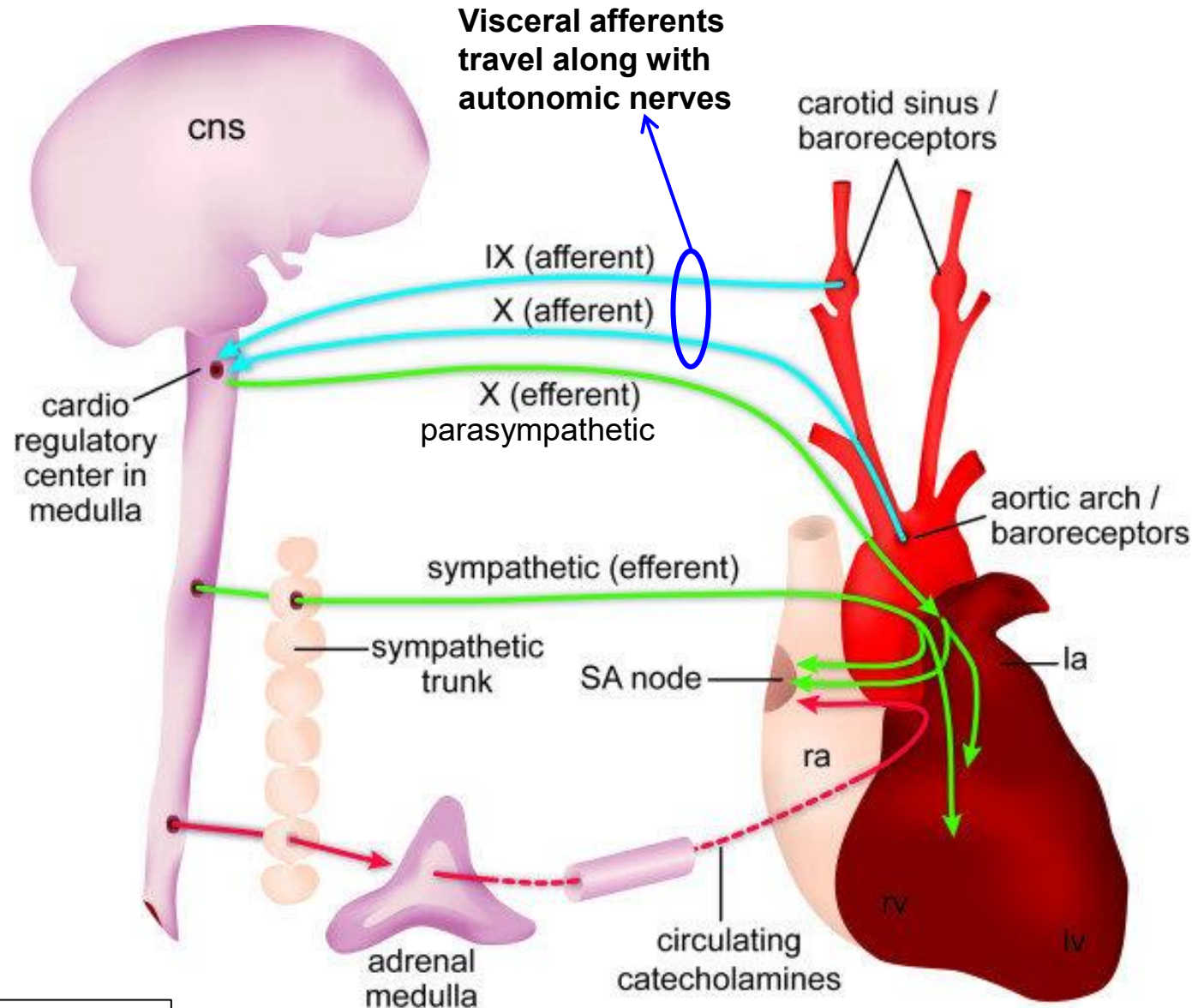
BP back to normal
(**Effect**)



Arterial Baroreceptors regulate blood pressure (BP)



Arterial Baroreceptors regulate blood pressure (BP)

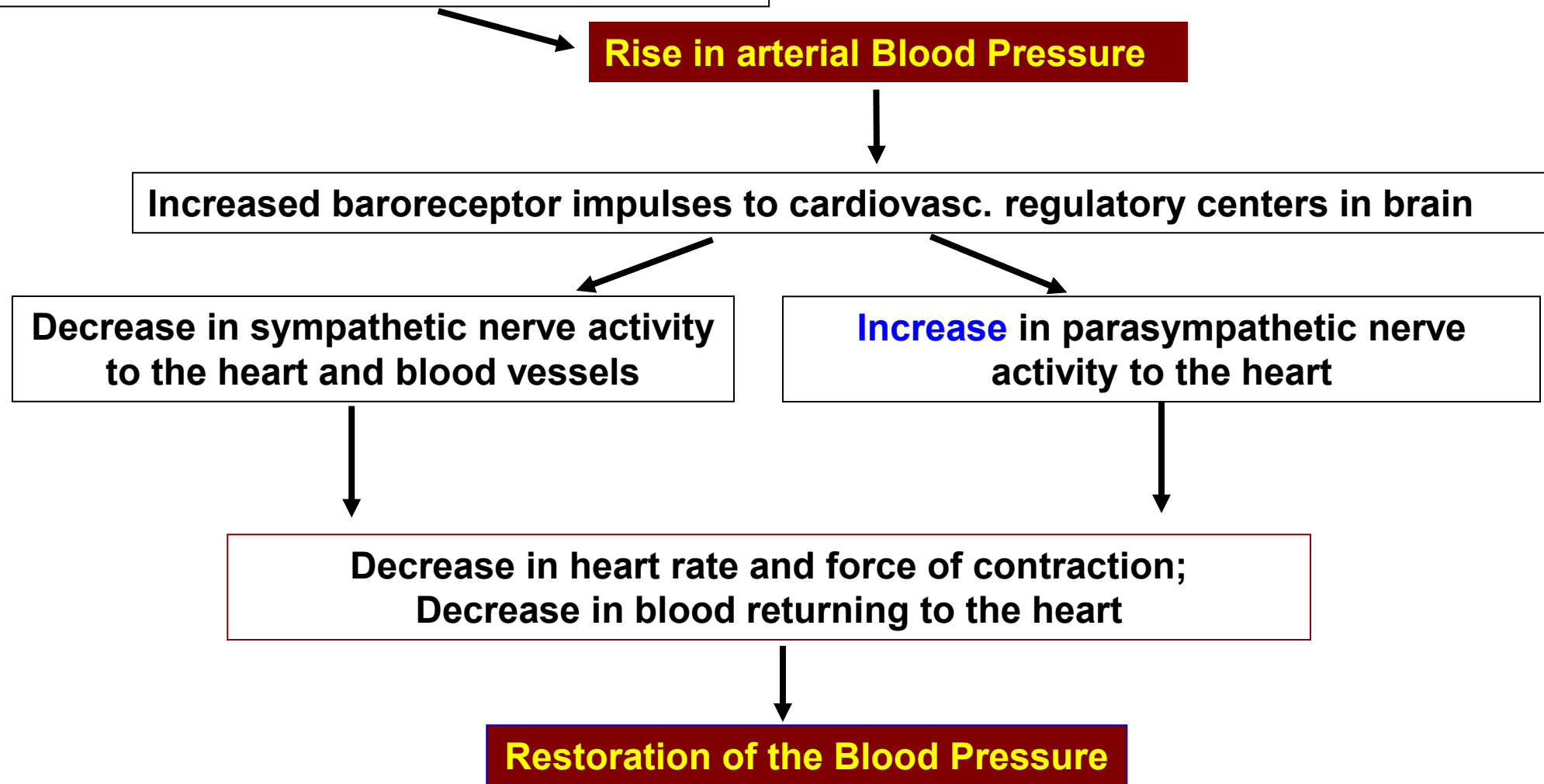


Baroreceptors
are sensors
located in carotid
sinus and aortic
arch

Blue: Afferent
nerves;
Green: Efferent
nerves;
Red: Efferent nerve
supplying adrenal
medulla

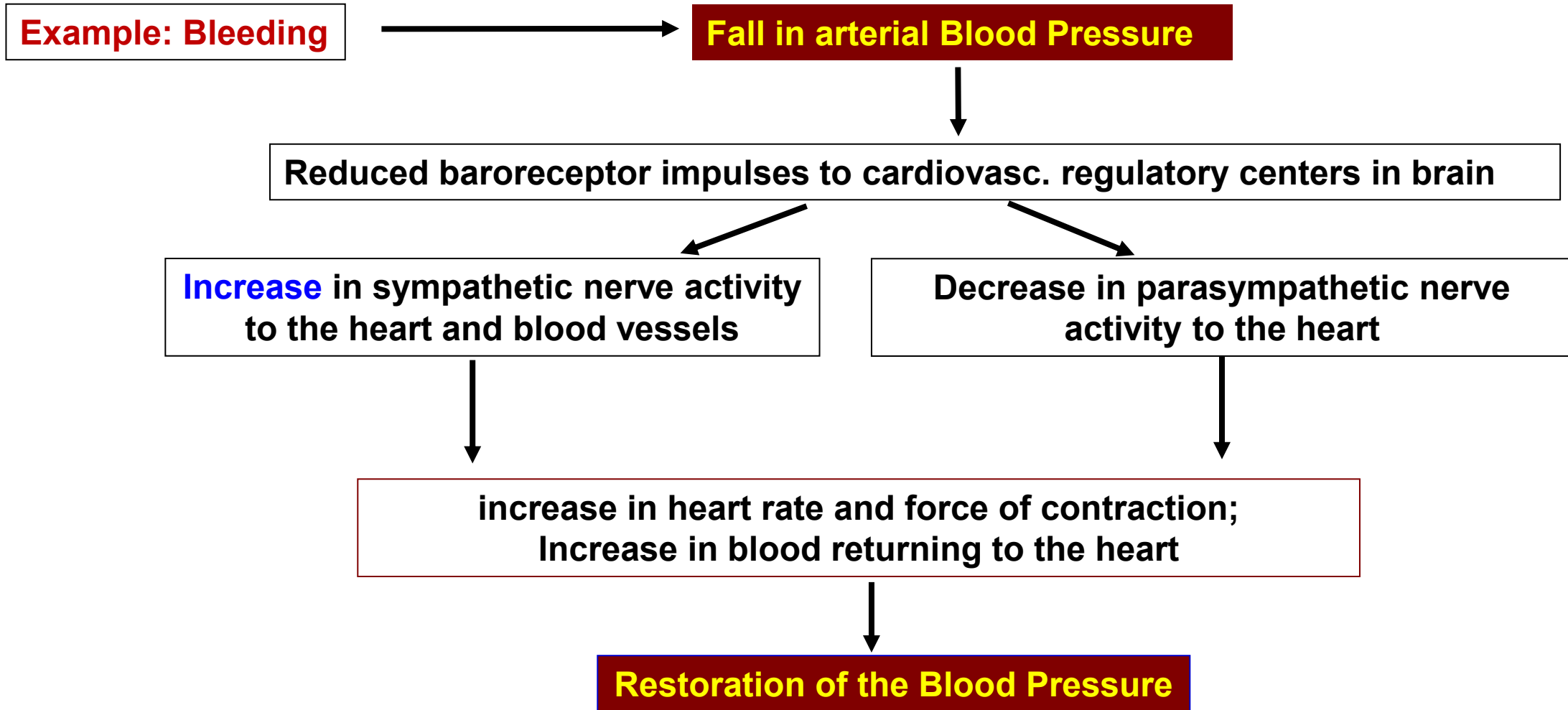
Baroreflex response to a sudden rise in BP

Example: Change posture from standing to supine



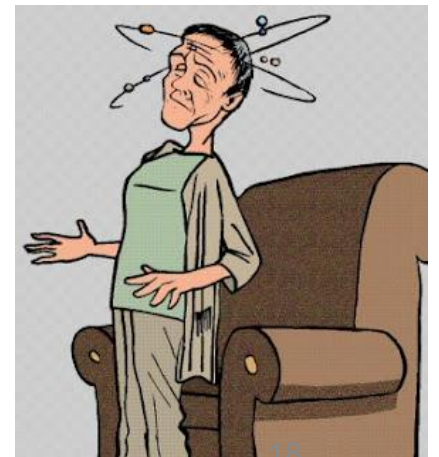
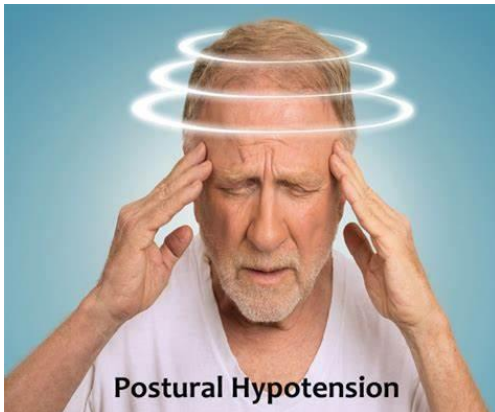
Task: Draw a similar flow chart to show baroreflex response to a **sudden decrease** in the Blood Pressure

Baroreflex response to a sudden lowering BP (Bleeding)



Why baroreflex is important in day-to-day life?

- ❖ Patient with dysfunction of the baroreflex is unable to maintain normal blood pressure under different circumstances.
- ❖ This leads to a low blood pressure while changing body posture from supine to erect and is called **orthostatic hypotension** (postural hypotension)
- ❖ Patient complaints:
 - Lightheadedness
 - Blurry vision
 - Vertigo/dizziness
 - Transient loss of consciousness (syncope)



An 82-year-old woman comes to the physician because of a 6-month history of lightheadedness and visual blurring while standing off the bed. She is a vegan and was diagnosed with peripheral neuropathy due to vitamin B12 deficiency six months ago. Her blood pressure in supine is 132/84 mm Hg and in standing is 110/70 mm Hg. She is suspected to have a dysfunction of the baroreflex. Which of the following structures most likely functions as effector organ in this reflex?

- A. Vagus nerve
- B. Baroreceptors
- C. Sympathetic nerves
- ✓ D. Heart
- E. Lungs
- F. Cardiovascular regulatory center



Any Questions ?

How to reach me: Either walk into my office or email.

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